Animated version

Examples of use of MIMOSA, a probabilistic framework for functional data analysis

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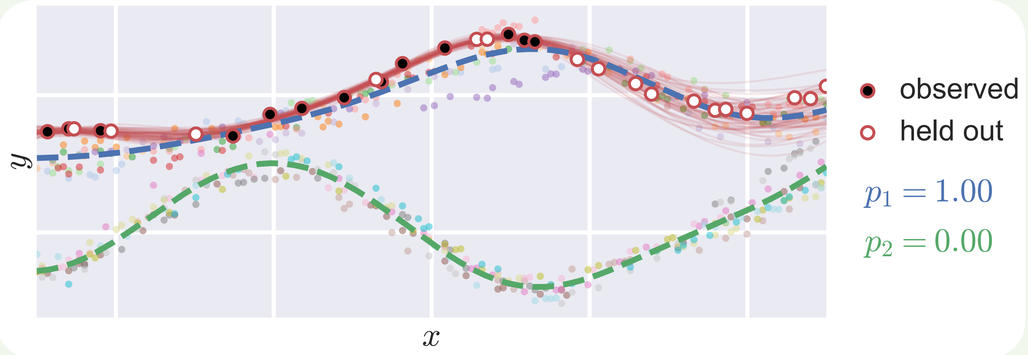
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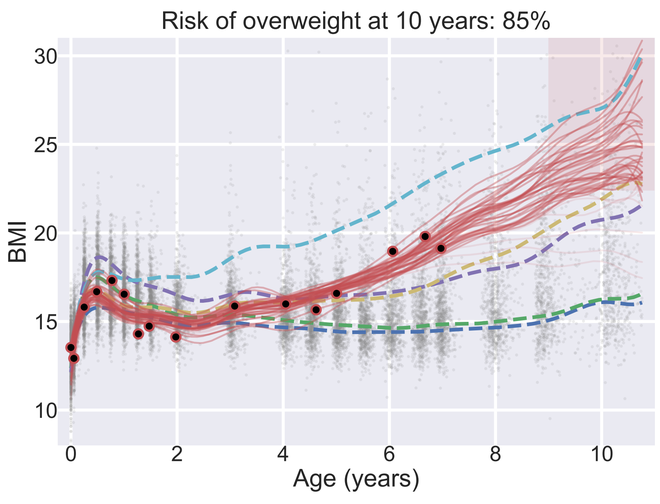
Intuition: a mixture of multi-task Gaussian processes

Each sample	=	cluster-specific mean trend	+	sample-specific variation	+	noise
$y_i(x)$	=	$\Sigma_k Z_{ik} \mu_k(x)$	+	$f_i(x)$	+	$\varepsilon_i(x)$

uncertainty aware · explainable · scalable · Bayesian · customizable

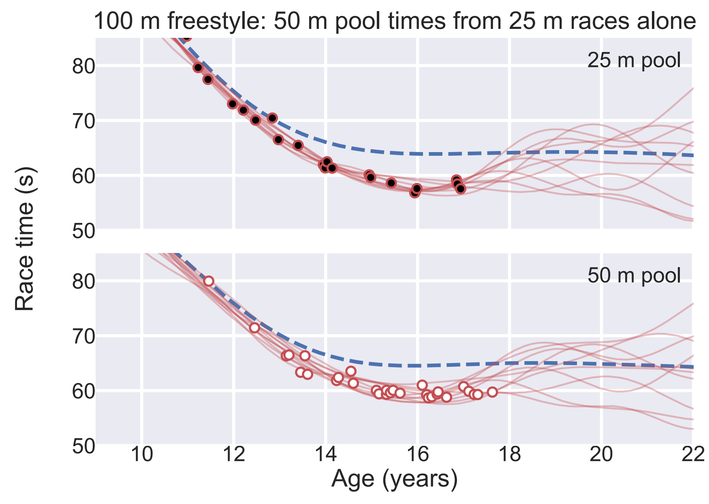


A · Prospective prediction — child BMI



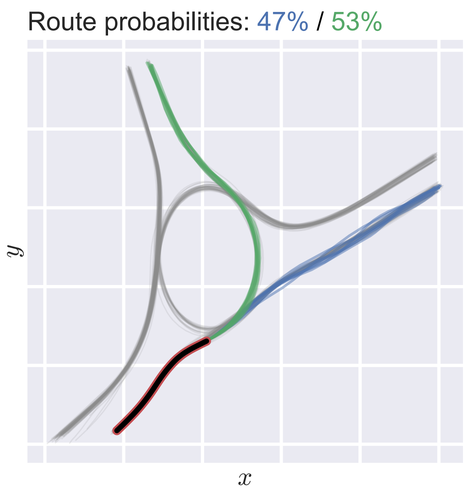
Estimates a child's BMI trajectory and risk of overweight at 10 years from a few early measurements and trends learnt from other children.

B · Multi-output prediction — swimming



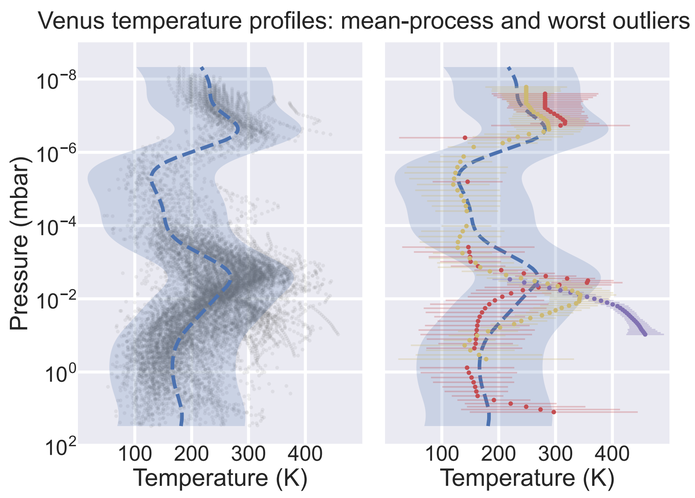
Predicts a swimmer's 50 m pool race times from her 25 m pool races alone, using inter-output correlations learnt from other swimmers.

C · Trajectory clustering — roundabout



Forecasts a car's trajectory through a roundabout from its first observations, with two likely routes extracted from other cars.

D · Outlier detection — Venus profiles



Detects outliers among temperature profiles measured on Venus by comparing each sample with its mean process.

MIMOSA opens up new possibilities in functional data analysis

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